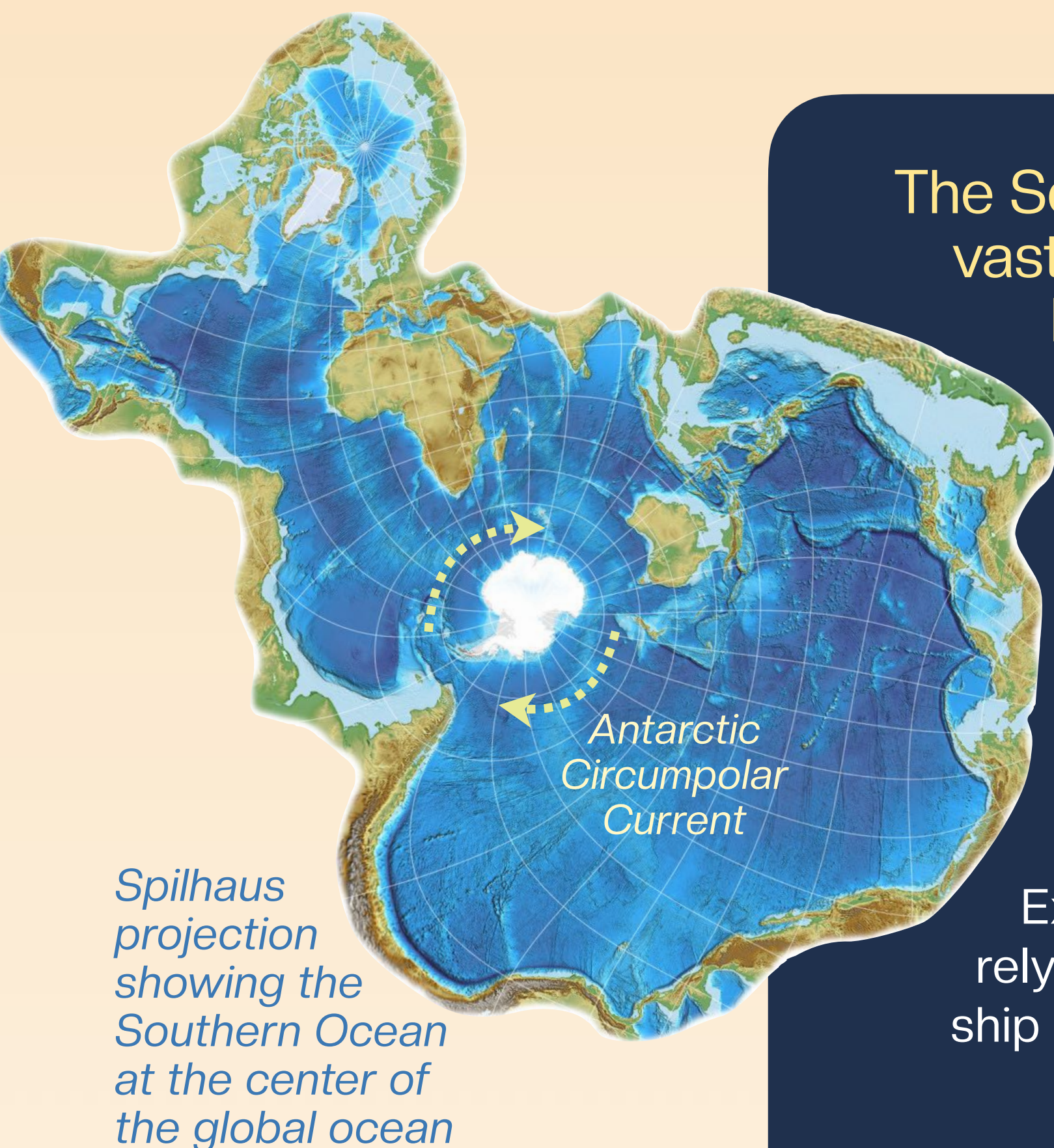


Filling in the gaps: carbon mapping in the data-sparse Southern Ocean

Machine learning reconstruction of ocean surface carbon using observations from ships and floats

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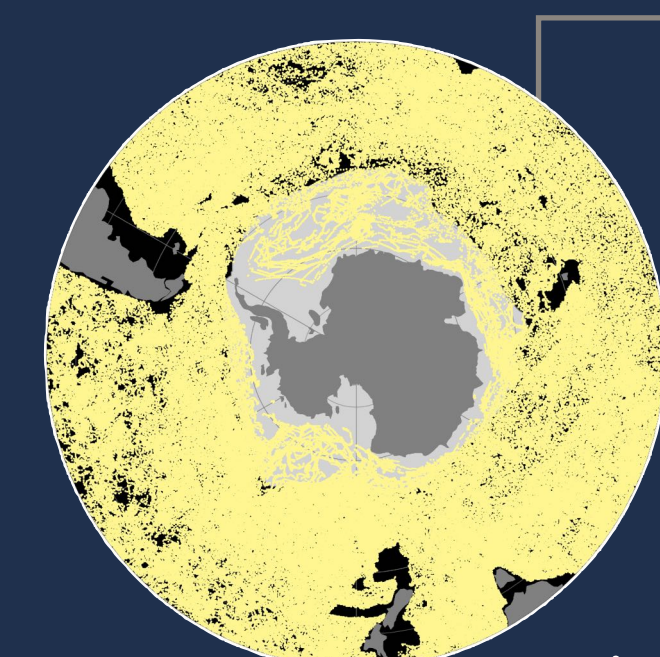
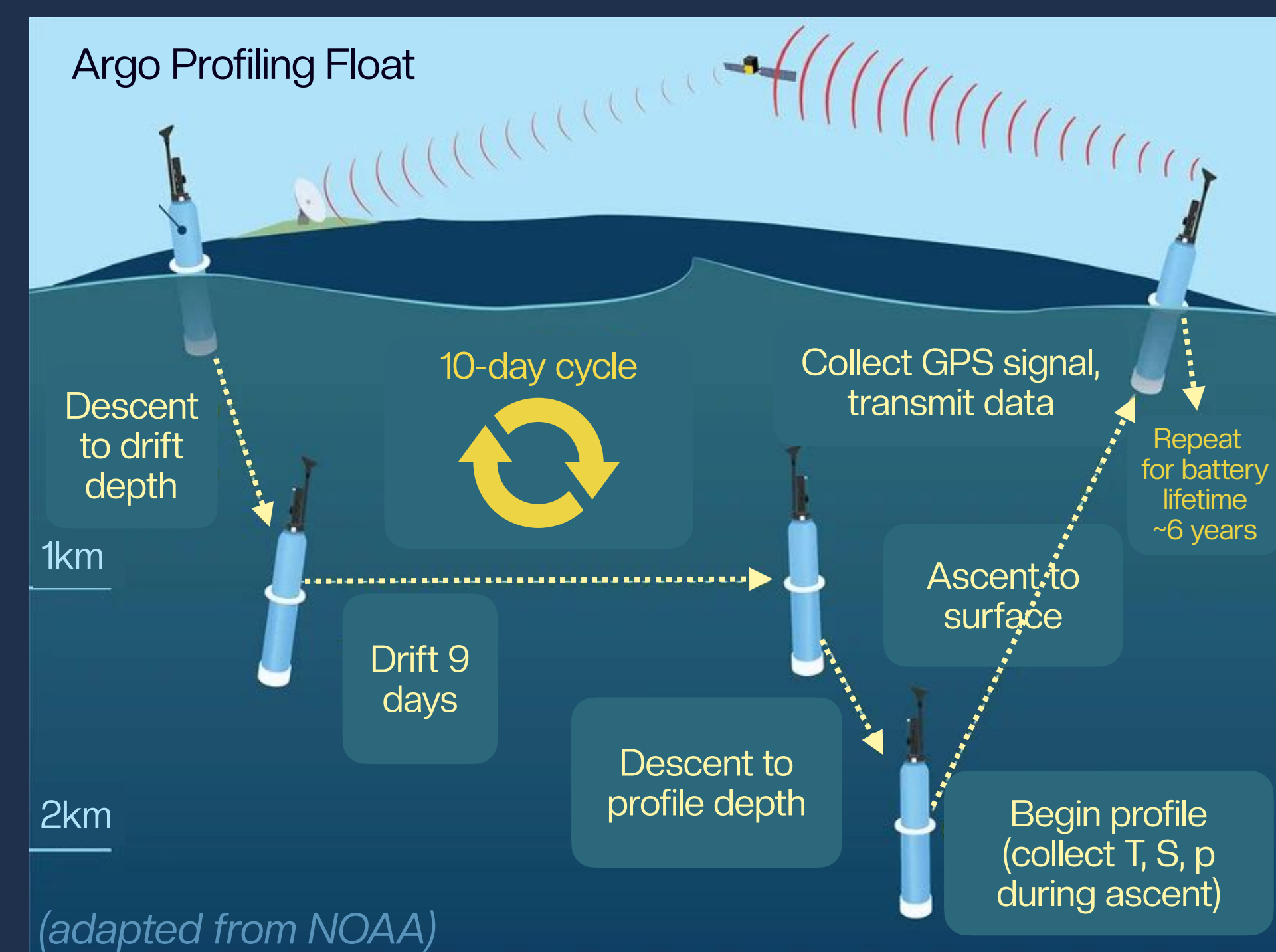
The Southern Ocean is a central hub where vast amounts of carbon are exchanged between the ocean and atmosphere.

The Southern Ocean, defined south of 35°S, is thought to account for **40% of the total ocean uptake of anthropogenic carbon**²

Large uncertainties in the Southern Ocean carbon sink contribute significantly to uncertainty in the global carbon cycle

Existing methods for surface pCO₂ mapping rely on monthly-averaged, spatially smoothed ship data (SOCAT) with **large observing gaps**

Autonomous instruments called **Argo floats** have dramatically increased observational coverage, including in the Southern Ocean.



Core-Argo (2001-) profiles of temperature and salinity; 2014–2023 shown

Biogeochemical (BGC)-Argo floats (2014–) measure additional properties including nitrate, oxygen, chlorophyll, pH

Together, BGC-Argo and SOCAT provide sufficient coverage to enable a new pCO₂ regression framework that avoids typical smoothing of the training data

Abbreviations:
Surface Ocean CO₂ Atlas (SOCAT)
Biogeochemical-Argo (BGC-Argo)
Partial pressure of CO₂ (pCO₂)

1. Breiman (2001); Mach. Learning, 2. DeVries et al. (2014); Glob. Biogeochem. Cycles, 3. Jones et al. (2023); Ocean Sci, 4. Park et al. (2023); Ann. App. Stat.

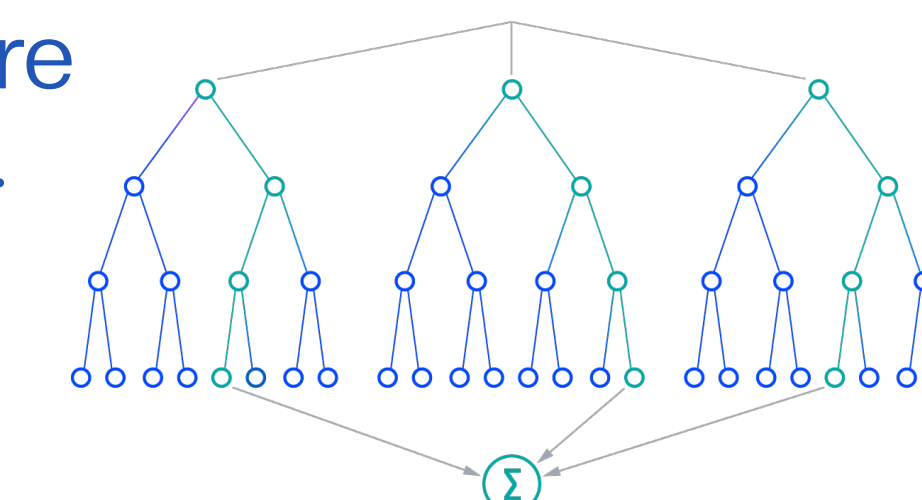
MODEL DEVELOPMENT

A multi-platform dataset is used to train **machine learning algorithms that estimate ocean surface carbon (pCO₂)** from more readily available measured properties.

Float and ship observations are organized into clusters using subsurface temperature, salinity (gaussian mixture modeling³)

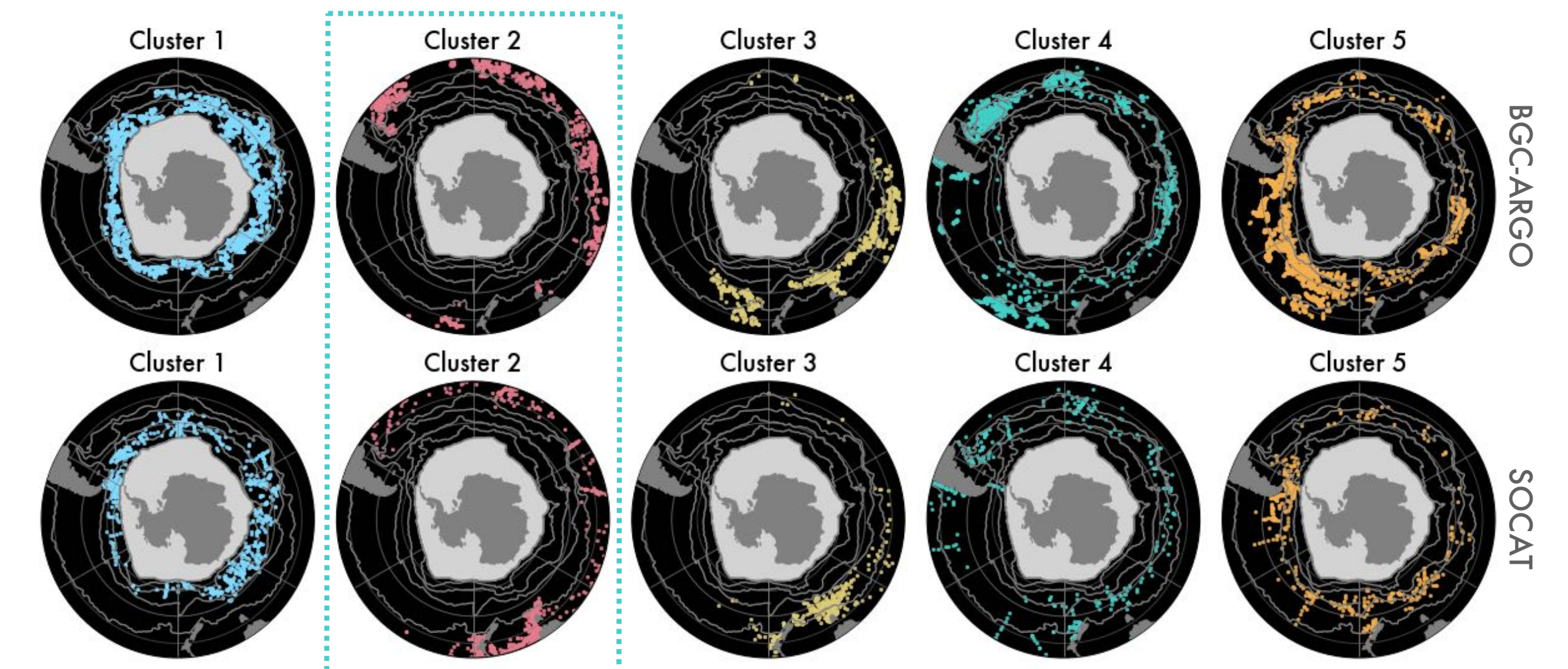
A separate regression model is trained for each cluster to learn the locally relevant mapping from available predictors to pCO₂ (random forest regression¹)

Using point data rather than monthly averages to learn underlying relationships is a key advance, especially in the turbulent Southern Ocean where properties can change rapidly and intensely



CRUSOE: Class-conditional Regression Unifying in Situ observations for Ocean Estimation

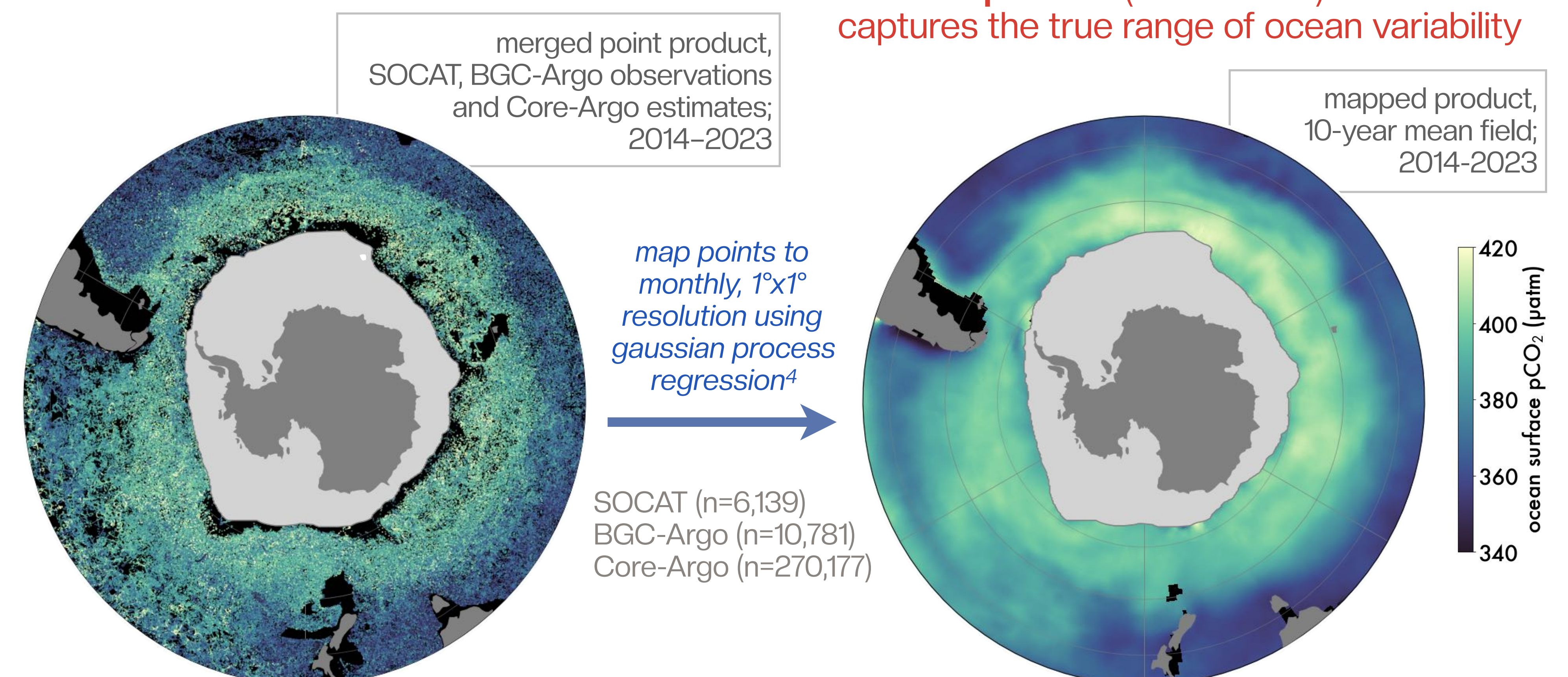
predictor variables:
temperature, salinity (floats)
mixed layer depth (floats)
sea surface height, sea ice (satellite)
wind speed (atmospheric reanalysis)
solar radiation (computed)



APPLICATION & IMPACT

Applying our model to widespread Core Argo temperature + salinity profiles **expands the Southern Ocean point pCO₂ dataset by 20-fold...**

and enables a **novel mapped surface carbon product (2014–2023)** that better captures the true range of ocean variability



Utilizing data from Core-Argo floats as model inputs for pCO₂ estimation **extends the value of these instruments beyond their original sensor array**

Our gridded product provides a key missing piece for better quantification of the oceanic carbon sink; reducing Southern Ocean uncertainty will be essential for constraining the Global Carbon Budget